

Final CNN comparison table: mean +/- std across seeds

Experiment	WD fixed	AdaDecay	OUIDecay
EfficientNet-B0 + Stanford Cars, ADAM, wd=1e-5	1.2676 +/- 0.0333	1.3397 +/- 0.0140	1.1667 +/- 0.0213
EfficientNet-B0 + Stanford Cars, ADAM, wd=5e-5	1.2019 +/- 0.0292	1.2543 +/- 0.0104	1.0594 +/- 0.0213
ResNet50 + Food101, ADAMW, wd=0.01	1.0333 +/- 0.0073	1.1204 +/- 0.0252	1.0336 +/- 0.0030
ResNet50 + Food101, ADAMW, wd=0.05	1.0241 +/- 0.0092	1.5827 +/- 0.1559	0.9976 +/- 0.0146
DenseNet121 + CIFAR100, ADAMW, wd=0.05	1.3703 +/- 0.0058	1.1710 +/- 0.0079	1.3337 +/- 0.0056
DenseNet121 + CIFAR100, ADAMW, wd=0.1	1.3465 +/- 0.0074	1.2878 +/- 0.0032	1.2752 +/- 0.0019
MobileNetV2 + CIFAR10, ADAM, wd=1e-4	0.3884 +/- 0.0033	0.4151 +/- 0.0000	0.3622 +/- 0.0044
MobileNetV2 + CIFAR10, ADAM, wd=5e-4	0.3487 +/- 0.0036	0.3892 +/- 0.0000	0.3298 +/- 0.0040

Rows show the 8 selected experiment settings. The AdaDecay column is AdaDecayG for EfficientNet and MobileNet, and AdaDecay for ResNet and DenseNet. Bold marks the best mean in each row.

VIP final table: reproducibility details

Reproducibility notes

Metric: mean +/- std of best_val_loss across seeds 1, 2, and 3, aggregated from descargas/results/*.pkl by the metadata fields model, dataset, optimizer, storage_method, weight_decay, and seed.
The runs were selected from the final CNN sweep and use the exact training code found in the corresponding python files, with the same augmentation, schedule, and clipping logic as the executed launchers.

EfficientNet-B0 + Stanford Cars (Adam, wd=1e-5 and 5e-5, AdaDecayG): the script trains for 100 epochs with batch size 64, base lr 8e-4, min lr 1e-5, 8 workers, label smoothing 0.05, and grad clipping at 1.0. The training transform uses RandomResizedCrop(224, scale=0.7-1.0, ratio=0.85-1.15), RandomHorizontalFlip, ColorJitter, RandAugment, and RandomErasing; validation uses Resize(256) and CenterCrop(224). Learning rate follows warmup plus cosine decay. No mixup or cutmix is used.

ResNet50 + Food101 (AdamW, wd=1e-2 and 5e-2, AdaDecay): the script trains for 50 epochs with batch size 128, base lr 3e-4, min lr 3e-5, and no explicit grad clipping. Training uses RandomResizedCrop(224, scale=0.6-1.0), RandomHorizontalFlip, ColorJitter, and standard normalization; validation uses Resize(256) and CenterCrop(224). The optimizer follows the same warmup plus cosine schedule used in the code, and the launcher keeps the dataset local under the repository datasets directory.

DenseNet121 + CIFAR100 (AdamW, wd=5e-2 and 1e-1, AdaDecay): the script trains for 100 epochs with batch size 256, base lr 5e-4, min lr 5e-6, fused AdamW enabled in auto mode, and no grad clipping. Training uses RandomCrop(32, padding=4), RandomHorizontalFlip, RandAugment(num_ops=2, magnitude=9), and RandomErasing; validation uses plain tensor conversion and normalization. The learning rate again uses warmup and cosine decay.

MobileNetV2 + CIFAR10 (Adam, wd=1e-4 and 5e-4, AdaDecayG): the script trains for 100 epochs with batch size 256, base lr 5e-4, min lr 5e-6, fused Adam enabled in the launcher configuration, and no grad clipping. Training uses the same CIFAR-style augmentation as DenseNet, namely RandomCrop(32, padding=4), RandomHorizontalFlip, RandAugment(num_ops=2, magnitude=9), and RandomErasing; validation is normalization only. The learning-rate schedule is warmup plus cosine decay.

- Columns:
- WD fixed = constant weight decay baseline.
 - AdaDecay = AdaDecay or AdaDecayG, depending on the row above.
 - OUIDecay = the original OUIDecay scheduler.